SAP PROJECT DOCUMENTATION

Procure-to-Pay (P2P) — Full Purchasing Cycle in SAP MM

ZTECH Industries Pvt. Ltd. | Company Code: ZT01 | Plant: ZP01 — Bhubaneswar

Interactive Simulation: https://p2pdashboardgit-sap-project.streamlit.app/

Github Link: https://github.com/Mou4338/p2p_dashboard.git

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Figure: High-Level Procure-to-Pay (P2P) Process Flow

Purchase Requisition *(ME51N)*	Request for Quotation *(ME41)*	Vendor Evaluation *(ME49)*	**Purchase Order** *(ME21N)*	Goods Receipt *(MIGO)*	Invoice Verification *(MIRO)*	Payment *(F110)*

Note: Purchase Order (ME21N) is the central binding document that anchors the entire P2P chain.

1. Introduction

- **SAP as an ERP Platform:** SAP (Systems, Applications, and Products in Data Processing) is one of the most widely deployed Enterprise Resource Planning platforms in the world. It brings every department of an organisation — procurement, finance, inventory, logistics, and HR — into a single connected system where transactions are posted the moment they happen. There is no waiting for data to travel between departments manually. Decisions are made on current, accurate information, and errors caused by re-entering the same data in multiple places are eliminated entirely.
- **Why Manufacturing Companies Need ERP in Procurement:** A mid-sized manufacturer like ZTECH raises hundreds of purchase transactions every week. Managing vendor communications, routing purchase approvals, updating stock records, and posting financial entries across disconnected systems is not just slow — it is fundamentally unreliable. SAP solves this by providing one controlled environment where every step of procurement feeds directly into finance and inventory without anyone having to manually carry data across.
- **What the Procure-to-Pay Process Covers:** The Procure-to-Pay (P2P) cycle is the full journey from recognising a supply need inside the organisation through to paying the vendor who fulfilled it. In SAP MM, this journey is represented as a chain of linked documents — Purchase Requisition, RFQ, Purchase Order, Goods Receipt, Invoice, and Payment — each produced at a specific step and tied back to the one before it. The chain is complete, traceable, and entirely system-driven.

- **Why Getting P2P Right Matters:** A well-configured P2P cycle keeps production lines running with materials arriving on schedule, keeps finance accurate with payments going out only for goods actually received, and keeps vendor relationships healthy by paying on time and within agreed terms. The three-way matching feature that SAP enforces at the invoice verification step is particularly important — it means no payment can go out unless a Purchase Order and a Goods Receipt both exist to support it.

1.1 Problem Statement

- In most traditional procurement setups, purchase requests travel on paper and get tracked through spreadsheets. The result is processing delays of several business days per request, with no visibility for managers who need to know when materials will arrive.
- Without a shared, structured vendor database, procurement teams have no way to compare supplier prices side by side, track historical delivery performance, or enforce pricing agreements from one order to the next.
- When goods receipts are recorded outside the main system, the physical stock count and the system stock count drift apart. This causes duplicate ordering of materials already in the warehouse, or stock-outs because the system shows stock that physically no longer exists.
- Manual three-way matching — sitting down with the PO, the delivery note, and the vendor invoice to check them line by line — is slow and misses things. Discrepancies that slip through produce duplicate payments and show up as findings in audits.
- Multi-level purchase approvals on paper create bottlenecks that are felt most painfully when materials are urgently needed for production, because every hour spent waiting for a signature is a production delay.
- SAP's integrated P2P cycle resolves all of these issues by digitising the entire procurement workflow within a single governed system where every approval is tracked, every document is linked, and every financial entry is posted automatically.

2. Company Overview — ZTECH Industries Pvt. Ltd.

Attribute	Details	Attribute	Details
Company Code	ZT01	**Industry**	Manufacturing & Distribution
Headquarters	Bhubaneswar, Odisha	**Company Size**	Mid-sized (fictitious scenario)
Plant	ZP01 — Bhubaneswar	**SAP System**	SAP ECC 6.0 / S/4HANA Compatible
Purchase Org	ZPO1 — Central Purchasing	**Currency**	Indian Rupee (INR)
Purchase Grp	ZPG — Procurement Team	**Storage Locs**	ZSL1 (Raw Materials) / ZSL2 (Finished Goods)
Scenario	500 KG Raw Steel Sheets	**PO Value**	INR 2,25,000 excl. tax | INR 2,65,500 incl. GST 18%

ZTECH Industries Pvt. Ltd. is a fictitious mid-sized manufacturing company operating out of Bhubaneswar, India. The company produces industrial equipment and structural components for clients across the country and sources its raw materials, engineering parts, consumables, and safety supplies from a network of registered vendors. The daily volume and complexity of procurement activity means an ERP-backed purchasing process is not optional for ZTECH — it is the only way to keep production on schedule and financial records clean.

All purchasing decisions for ZTECH flow through a centralised procurement model under Purchase Organisation ZPO1. The main plant, ZP01 in Bhubaneswar, draws materials from two storage locations: ZSL1 for incoming raw materials and ZSL2 for finished goods held before despatch. No vendor commitment can be made without clearing the release strategy that SAP enforces on every requisition and purchase order.

3. Tech Stack & SAP Module Coverage

Layer	Tool / Module	Purpose
Core ERP	**SAP ECC 6.0 / S/4HANA**	Central integrated platform — MM, FI, SD, CO, ABAP
Procurement	**SAP MM**	PR, RFQ, Purchase Order, Goods Receipt, Invoice Verification
Financial Accounting	**SAP FI / FICO**	GR/IR Clearing, Vendor Payable, Bank Clearing, Tax Postings
Sales Integration	**SAP SD**	Vendor/Customer master alignment and Order-to-Cash linkage
Programming Layer	**SAP ABAP**	Custom reports, SmartForms for PO output, release-strategy exits
Simulation Tool	**Python + Streamlit**	Interactive 7-step P2P simulation dashboard with live scoring
Statutory Compliance	**GST / TDS**	Indian tax codes (GST 18%) configured in PO pricing and MIRO

4. Procure-to-Pay (P2P) — Detailed End-to-End Process

The P2P cycle is the operational backbone of ZTECH's procurement function. Every business event — a material request raised by production, a vendor's delivery arriving at the gate, or a Finance team member verifying an invoice — is captured in SAP as a linked document the moment it occurs. The result is an unbroken procurement chain from the very first internal request all the way through to the final payment, with a clear audit trail at every step.

4.1 Purchase Requisition

- **T-Code:** ME51N
- **Tables:** EBAN (Purchase Requisition Header and Item Data)
- **How It Works:** Everything in the P2P cycle begins with a Purchase Requisition. When the Production department identifies that 500 KG of Raw Steel Sheets are needed, a team member opens ME51N in SAP and fills in the material number, required quantity, unit of measure, requested delivery date, and the cost centre (CC-PROD-01) against which the spend will be charged. Once saved, the PR enters the release workflow. An authorised approver must mark it Released before the procurement team can take

any action — the system blocks all further steps until that happens. This release gate is what keeps spending controlled from the very first document in the chain.

- **Business Value:** The Purchase Requisition turns a verbal or informal supply need into a formal, system-tracked demand with budget accountability attached. From the moment it exists in SAP, the cost centre is committed, the requirement is visible to procurement, and the purchasing team has a documented starting point for the entire cycle.

4.2 Request for Quotation (RFQ)

- **T-Codes:** ME41 (Create RFQ) | ME47 (Maintain Quotation)
- **Tables:** EKKO (Purchasing Document Header), EKPO (Item Data)
- **How It Works:** With the PR approved and released, the procurement team sends a formal Request for Quotation to a shortlist of vendors registered in the SAP vendor master using ME41. The RFQ communicates exactly what ZTECH needs — the material specification, the required quantity, the expected delivery date, and the deadline for returning a price. When vendor responses come in, the team enters each supplier's unit price and delivery commitment into SAP via ME47. SAP immediately collates all responses and builds a price comparison list in the background, ready for the evaluation step.
- **Business Value:** Recording all vendor quotations inside SAP means every procurement decision rests on documented, retrievable data rather than informal phone calls or emails. The full history of who quoted what price on which date is always available — for internal review, compliance checks, or external audits.

4.3 Vendor Evaluation and Selection

- **T-Codes:** ME49 (Price Comparison List) | ME61 (Vendor Evaluation)
- **Tables:** EKKO, EKPO, LFM1 (Vendor Master Purchasing Data)
- **How It Works:** ME49 produces a ranked comparison of all vendor quotations by total order value. For this project, a composite scoring model is layered on top: unit price carries 40% of the weight, delivery speed accounts for 30%, and the vendor's quality history covers the remaining 30%. The supplier with the strongest composite score is recommended for the Purchase Order. Vendors who are not selected receive formal rejection notifications generated from SAP. The full scoring simulation is available at: https://p2pdashboardgit-sap-project.streamlit.app/
- **Business Value:** Composite scoring removes the temptation to simply award orders to whoever quoted the lowest price. It forces procurement to weigh delivery reliability and quality track record alongside cost, reducing the operational risk that comes from chasing price at the expense of performance.

4.4 Purchase Order Creation

- **T-Code:** ME21N
- **Tables:** EKKO (PO Header), EKPO (PO Item), EKET (Delivery Schedule Line)
- **How It Works:** With a vendor chosen, the procurement team creates the Purchase Order through ME21N. The PO spells out every detail the vendor needs to fulfil the order: material description, quantity (500 KG), agreed unit price (INR 450/KG), required delivery date, destination storage location (ZSL1), and payment terms. Before the document reaches the vendor, it passes through ZTECH's release strategy — at INR 2,25,000 this falls under the Department Manager approval tier. Once released, SAP transmits the PO to the vendor as the legally binding procurement instruction.

- **Accounting Note:** No accounting entry is generated when the PO is created. However, SAP records the commitment in its commitment accounting layer, so Finance can immediately see the outstanding financial obligation against the approved budget — even before any goods arrive.
- **Business Value:** The Purchase Order replaces informal purchasing commitments with a documented, legally enforceable agreement. All subsequent documents — Goods Receipt, Invoice, Payment — reference the PO number, which is what makes end-to-end document flow traceability possible in SAP.

4.5 Goods Receipt

- **T-Code:** MIGO
- **Tables:** MKPF (Material Document Header), MSEG (Document Item), MARD (Stock Data by Storage Location)
- **How It Works:** When the vendor's truck pulls up at Plant ZP01, the warehouse team inspects the delivery against the open Purchase Order. Using MIGO in SAP, they confirm the received quantity (500 KG), assign it to Storage Location ZSL1, record the batch number, and capture the quality inspection status. SAP immediately posts a Material Document that updates stock levels, and simultaneously generates an Accounting Document in FI — no one in Finance has to be contacted for this to happen.
- **Inventory Impact:** The moment the GR is saved, the 500 KG of Raw Steel Sheets appears in ZSL1 as available unrestricted stock — or as quality stock if the material is flagged for quality inspection. Production can see the newly arrived material straight away. If it is under quality hold, it is not available for production orders until the QC team formally releases it.

Accounting Entry at Goods Receipt:

Dr	Stock Account (Raw Materials) — inventory balance increases to reflect goods physically received at plant
Cr	GR-IR Clearing Account — a bridging liability is parked here pending the vendor's invoice

- **Business Value:** The Goods Receipt is the document that arms SAP's three-way match. Invoice Verification in the next step cannot proceed — and no payment can be made — until a GR exists. This single control ensures ZTECH never pays for goods that have not physically arrived and been confirmed at the plant.

4.6 Logistics Invoice Verification

- **T-Code:** MIRO
- **Tables:** RBKP (Invoice Document Header), RSEG (Invoice Line Items), BKPF, BSEG
- **How It Works:** When the vendor's invoice lands with the Finance team, they enter it into SAP via MIRO. SAP instantly runs a three-way match, cross-checking the invoice against both the Purchase Order and the Goods Receipt. For this project the invoice covers 500 KG at INR 450/KG plus GST at 18%, giving a total of INR 2,65,500. If every figure sits within the configured tolerance — typically 5% variance — SAP posts the invoice and queues it for payment. Any value outside that tolerance triggers a system hold, and Finance contacts the vendor to resolve the discrepancy before anything moves to payment.

Accounting Entry at Invoice Posting:

Dr	GR-IR Clearing Account — the bridging liability parked at Goods Receipt is now cleared
Dr	Input Tax Account (GST 18%) — recoverable input tax is captured as a separate debit
Cr	Vendor Payable Account — the definitive liability to the vendor (INR 2,65,500) is now on the books

- **Business Value:** Automated three-way matching takes one of the most error-prone manual tasks in Finance and makes it instantaneous and reliable. Duplicate invoices are rejected at entry. Vendor overcharging is caught before payment. Accounts payable always reflects only invoices that are backed by physical proof of delivery.

4.7 Payment Processing

- **T-Codes:** F110 (Automatic Payment Run) | F-53 (Manual Outgoing Payment)
- **Tables:** BKPF (Accounting Document Header), BSEG (Line Items), PAYR (Payment Medium)
- **How It Works:** The Finance team runs the Automatic Payment Program through T-Code F110 on a regular schedule — typically weekly or fortnightly. SAP scans every open vendor invoice, identifies those due within the configured payment window, and processes them in one batch. For ZTECH's steel procurement, F110 selects the INR 2,65,500 invoice, creates a Payment Document and a Clearing Document, removes the open item from the vendor's account, and dispatches a bank transfer instruction to house bank HDFC Bank (ZB01). The whole run requires no manual bank handling.

Accounting Entry at Payment:

Dr	Vendor Payable Account — the liability to the vendor is fully cleared from ZTECH's books
Cr	Bank Outgoing Clearing Account — the cash outflow of INR 2,65,500 is recorded against the house bank

- **Business Value:** Running payments through F110 eliminates missed due dates, which protects vendor relationships and keeps ZTECH's credit standing intact. It also automatically identifies and captures any early-payment discount available under agreed terms — savings that a manual payment process would routinely miss.

With the F110 payment run complete, the Procure-to-Pay cycle for this procurement is fully closed. The vendor account is cleared to zero, every document in the chain — from the original PR through to the bank clearing entry — is linked in SAP's document flow, and the entire INR 2,65,500 transaction is recorded, reconciled, and audit-ready.

5. T-Code Summary

#	Process Step	T-Code	What It Does
1	**Purchase Requisition**	ME51N	Raise an internal material request on behalf of the department
2	**Request for Quotation**	ME41 / ME47	Publish RFQ to vendors and record their price responses

3	**Vendor Evaluation**	ME49 / ME61	Score and rank vendors on price, delivery lead time, and quality
4	**Purchase Order**	ME21N	Issue the approved, legally binding order to the chosen vendor
5	**Goods Receipt**	MIGO	Confirm delivery and update warehouse stock levels in real time
6	**Invoice Verification**	MIRO	Perform three-way match and post the verified vendor invoice
7	**Payment Processing**	F110 / F-53	Run automatic or manual payment and clear the vendor's open item

6. SAP Tables Reference

Step	Document Type	SAP Tables	Purpose
PR	**Requisition**	EBAN	Stores the purchase request header along with every line-item detail
RFQ / PO	**Purchasing Doc**	EKKO, EKPO, EKET	Covers purchasing document headers, item data, and delivery schedule lines
GR	**Material Doc**	MKPF, MSEG, MARD	Records goods movements and keeps storage-location stock balances current
Invoice	**Invoice Doc**	RBKP, RSEG	Holds vendor invoice header information and all individual line items
Accounting	**FI Document**	BKPF, BSEG	Captures every financial posting generated throughout the P2P cycle
Payment	**Payment Doc**	PAYR, BSIK, BSAK	Tracks open payables, cleared items, and payment media output to the bank

7. Accounting Impact — Financial Postings

Step	Journal Entry	Effect on the Books
Goods Receipt (MIGO)	Dr Stock Account / Cr GR-IR Clearing	Inventory value rises; a clearing liability is parked until the invoice arrives
Invoice Verification (MIRO)	Dr GR-IR Clearing + Dr Input Tax / Cr Vendor Payable	Clearing account wiped out; GST input tax captured; vendor payable crystallised
Payment (F110)	Dr Vendor Payable / Cr Bank Outgoing Clearing	Vendor liability fully settled; cash outflow hits the bank clearing account

8. Technology Stack and Integration of Modules

This project runs on the SAP ERP platform (ECC 6.0 / S/4HANA compatible), where the MM, FI, SD, and ABAP layers operate as one connected system. Alongside the core SAP configuration, a Python and Streamlit simulation dashboard was built to demonstrate all seven P2P steps interactively and in real time.

Interactive Dashboard: https://p2pdashboardgit-sap-project.streamlit.app/

Trigger Event	From	To	What Happens Automatically
Goods Receipt (MIGO)	MM	FI	Stock count updated instantly; accounting document auto-posted in FI
Invoice Verification (MIRO)	MM	FI	GR-IR Clearing wiped; Vendor Payable entry created without manual posting
Vendor Payment (F110)	FI	Bank	Vendor Payable cleared; bank transfer instruction sent to house bank ZB01
Goods Issue on Delivery	SD	MM / FI	COGS debited and Finished Goods Inventory credited automatically

- **SAP MM (Materials Management):** MM is the engine of the P2P cycle. It manages every procurement document from the Purchase Requisition through to the Goods Receipt, and it keeps ZTECH's stock levels at Plant ZP01, Storage Location ZSL1 accurate in real time throughout.
- **SAP FI (Financial Accounting):** FI receives automatic accounting entries from MM at every key step — stock valuation at Goods Receipt, vendor payable at Invoice Verification, and bank clearing at Payment. The general ledger always reflects the current state of procurement without the Finance team having to raise a single manual journal entry.
- **SAP SD — Integration:** Although SD is not the primary module in P2P, its vendor and customer master data structures align with MM's procurement master. This shared data layer ensures the same entity is not maintained separately in two places, preserving data consistency across all business processes.
- **SAP ABAP (Programming Layer):** ABAP provides the extensibility that makes SAP's standard processes fit ZTECH's specific needs — custom vendor scoring reports, professionally formatted PO output using SmartForms, and automated release-strategy notification messages to approvers.
- **Python + Streamlit Dashboard:** The simulation application, built using Python with Streamlit, Pandas, and Plotly, walks through all seven P2P steps live: raising a PR, comparing vendor quotes, scoring vendors by composite method, building a PO with GST tax calculation, posting a GR, running a three-way match, and executing a payment — all with SAP-accurate document numbers and accounting outputs.
- **Why Cross-Module Integration Is the Core Strength:** When a Goods Receipt is posted in MM, three things happen simultaneously without anyone pressing a second button: the stock record updates, an accounting document posts in FI, and the PO delivery history refreshes. No inter-department communication is needed. No lag exists. No discrepancy can develop between what procurement recorded and what Finance holds on the books.

9. Unique Points / Competitive Advantages

Module / Feature	Core Capability	Business Benefit
SAP MM	PR, RFQ, PO, Goods Receipt, Stock Management	Full procurement lifecycle under one controlled system

SAP FI	GR-IR, Vendor Payable, Bank Clearing, Input Tax	Every financial entry posted automatically — zero manual journals
SAP ABAP	Custom reports, SmartForms, Release-Strategy Exits	SAP standard processes extended to fit ZTECH's specific requirements
Streamlit Dashboard	7-step P2P simulation, vendor scoring, tax calc	All seven cycle steps demonstrated live with SAP-accurate outputs
Three-Way Match	PO × GR × Invoice cross-verification	Duplicate payments and vendor overcharges blocked at source

- **Real-Time MM–FI Integration:** Every Goods Receipt and Invoice Verification immediately triggers the corresponding financial posting in SAP FI. Standard procurement flows produce zero manual journal entries — reconciliation errors caused by delayed or missed postings are eliminated at the architectural level.
- **India-Specific Statutory Compliance:** GST tax codes for CGST, SGST, and IGST at 18% are configured directly in the Purchase Order pricing procedure and carried through to MIRO. Tax is calculated, captured, and reported correctly without anyone having to perform a separate manual tax entry.
- **Three-Way Match Enforcement:** SAP will not allow a vendor invoice to be approved for payment unless a matching Purchase Order and a confirmed Goods Receipt both exist. This single mechanism blocks duplicate payments and vendor overcharging at the point of invoice entry — before any money leaves the company.
- **Value-Based Release Strategy:** Every Purchase Requisition and Purchase Order is subject to a configured release strategy that routes it to the right approval authority based on value. No vendor commitment can be made on an unreleased document — the system enforces this without depending on anyone to remember the rule.
- **Interactive Simulation Dashboard:** The Streamlit dashboard reproduces all seven P2P steps live, complete with SAP-accurate document numbers, journal entry outputs, and a configurable vendor composite scoring model — making the project both technically sound and practically demonstrable.
- **Scalable Phase-2 Roadmap:** The next development phase covers SAP HR Payroll integration for employee cost allocation, SAP PP production order costing, and SAC Planning for procurement budget forecasting and rolling variance analysis.

10. Conclusion

- This SAP project has walked through the complete Procure-to-Pay business cycle for ZTECH Industries Pvt. Ltd. from start to finish — covering organisational configuration, all seven process steps from ME51N through F110, automatic financial postings, India-specific GST compliance, and a full interactive simulation. Every element works as part of an integrated chain, not as a standalone feature.
- Each problem raised at the outset of the project has a direct answer in the implementation: SAP's release strategy removes approval bottlenecks; composite vendor scoring replaces guesswork with documented, objective data; real-time GR posting eliminates the gap between physical and system stock; and automated three-way matching prevents the payment control failures that cost organisations money and audit credibility.
- The Streamlit simulation dashboard makes these concepts tangible and testable — all seven steps are walkable live, with accurate document outputs at each stage. The dashboard is at: https://p2pdashboardgit-sap-project.streamlit.app/

- Taken together, this project demonstrates that SAP is not simply a record-keeping tool. When properly designed and configured, it is the structural framework that makes procurement transparent, financially controlled, and operationally reliable — and that is the foundation any career in SAP consulting or development needs to be built on.

11. References

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